Multimedia Signal processing

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1. **Question (8 pts)**:
   Consider a discrete memoryless source X that emits symbols from the alphabet $\{a, b, c\}$ according to the following p.m.f.:

symbol	Pr.
a	0.5
b	0.4
c	0.1

   (a) Determine the entropy of the source and lower/upper bounds on the average code length of a Shannon code that encodes $N = 1$ symbols at the time.

   (b) Build a Huffman code for this source ($N = 1$) and compute its average code length. Compare such length with the bounds above.

   (c) Consider a code that encodes $N = 2$ symbols at the time. Revise the lower/upper bounds above, build a Huffman code and compare its length with the revised bounds.

   (d) Consider the following sequences of symbols: *bccb*. Compute the number of bits per symbol using the Huffman codes above ($N = 1, 2$) and compare it with the bounds. Comment the result.

2. **Question (5 pts)**
   Discuss the design of an open-loop vocoder.

3. **Question (5 pts)**
   (a) Illustrate the block diagram of a transform coder.

   (b) Describe how to determine the optimal bit allocation, once the transform is given.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**

   (a) Consider a DPCM coder.

   - [T] [F] $1 \leq G_{DPCM} < G_{VQ}$, where G_{DPCM} and G_{VQ} denote, respectively, the coding gain of DPCM and vector quantization.
   - [T] [F] The quantization error on the prediction residual is the same as the quantization error on the original samples.
   - [T] [F] The predictor is computed based on previous original samples, i.e., $\hat{x}(n) = f(x(n-1), x(n-2), \ldots)$
   - [T] [F] In an Adaptive DPCM coder (ADPCM), it is possible to vary both the quantizer and the predictor.

   (b) Consider an image coder based on transform coding

   - [T] [F] If $N > M$, then $R_N(D) \leq R_M(D)$, where $R_N(D)$ denotes the number of bits needed to encode the image at a target distortion D, when using $N \times N$ blocks.
   - [T] [F] The different color components in natural images (e.g., RGB components) are statistically independent.
   - [T] [F] When adopting threshold coding, each image block is allocated the same number of bits.

   (c) Consider a motion estimation/motion compensation algorithm.

   - [T] [F] Motion estimation is more computationally expensive than motion compensation.
   - [T] [F] When increasing the block size, the energy of the prediction residuals increases.
   - [T] [F] Motion estimation/compensation at sub-pixel accuracy requires to interpolate both the current and the reference frame.

5. **Question (Matlab - 10pts)**

   Load the files $f1.mat$ and $f2.mat$ which contain two consecutive frames of a video sequence.

   (a) Encode $f2$ with a PCM encoder. Use the following values for the step size Q = [2 4 8 16 32 64]. Plot the rate-distortion function, where rate is estimated based on the entropy of the quantized symbols, and distortion is measured through the PSNR.

   (b) Encode $f2$ with a *closed-loop* DPCM encoder. Use the same step sizes as before. Plot the rate-distortion function in this case. Comment the result.